

Stanley Oduor Omondi

EDUCATION

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Columbia University

New York City, NY

B.A in Computer Science

August 2022 – May 2026

Courses taken: Introduction to Programming in Java, Data Structures and Algorithms, Discrete Mathematics, Linear Algebra and Probability, Advanced Programming, Introduction to Databases, Artificial Intelligence, Natural Language Processing, Cloud Computing, User Interface Design, Fullstack Web Development

SKILLS

Technical Skills: Java, Python, C, SQL, MySQL, HTML/CSS, Javascript, Typescript, Bash

Libraries, Frameworks and Tools: Unix/Linux, Sockets API, TCP/IP, HTTP/HTTPS, React.js, Node.js, FastAPI, OpenGL, CI/CD, PyTorch, TorchVision, , Git, GCP

EXPERIENCE

Solutions Engineer

May 2025 - August 2025

Oracle

Redwood Shores, CA

- Designed and implemented Oracle Human Capital Management(HCM) Fast Formulas for payroll processes and absence management using PL/SQL, addressing challenges in input validation, output debugging and edge-case handling tailored to client requirements.

- Designed, implemented and deployed AI agents within Oracle Fusion HCM to simplify the development and debugging of Fast Formulas and enhance understanding of HCM processes, configurations and results.

- Engaged with clients to evaluate adoption of Oracle Redwood and AI functionalities within Oracle HCM and contributed to regression testing and script validation for complex configurations with silent errors.

Operations Assistant(On-campus student employment)

February 2023 - April 2025

Columbia University IT department

New York, NY

- Led campus-wide computer maintenance and support a user base of over 500 students and faculty, ensuring maximum uptime and minimal downtime.

- Deployed cutting-edge hardware and software systems for hybrid class instructions such as interactive displays and wireless presentation systems, leveraging skills in virtualization and database management to ensure seamless technology

- Collaborated with a team of three students and full time employees in conducting inventory and upgrading the systems in the CUIT hyflex classrooms.

PROJECTS

LSTM for image recognition

March 2025 – June 2025

Natural Language Processing

Python, PyTorch, TorchVision, Python Image Library

- Built a complete PyTorch pipeline for data preprocessing, model training, and evaluation on the Flickr8k dataset, achieving strong caption generation performance.

- Implemented an image captioning model combining a pretrained ResNet-18 CNN encoder for visual feature extraction with a conditioned LSTM decoder to generate natural language captions for Flickr8k images.

- Enhanced training efficiency by precomputing CNN features and implemented advanced sequence decoding strategies including greedy, sampling, and beam search to improve caption quality.

2048 Game AI

Sept 2024 - Dec 2024

AI Agent project - Intelligent Search and Heuristic Evaluation

Python

- Developed an AI-driven decision engine in Python leveraging iterative-deepening expectiminimax with alpha-beta pruning, probabilistic modeling, and time-bound search optimization.

- Designed a complete game simulation environment in Python with modular components for game state management, move logic, AI decision-making, and real-time performance tracking.

HTTP client and server

Sept 2023 - Dec 2023

Network Programming

C, TCP/IP, HTTP, POSIX sockets

- Built a complete client-server architecture in C implementing an HTTP web server, custom HTTP client, and TCP-based backend lookup service, applying socket programming, DNS resolution, and secure file handling for reliable network communication.

- Integrated static and dynamic content delivery by hosting HTML/images and processing live database queries through persistent backend connections, with robust request validation, error handling, and end-to-end logging aligned to industry best practices.

AWARDS AND HONORS

Science Research Fellow, Columbia University

April 2022 - May 2026

Matt Pincus Scholarship Fund, Columbia University

January 2023, January 2024

LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

Secretary Columbia University ACF, Advent Collegiate

Africa Development Group

National Society of Black Engineers